

Determinant of a Matrix

$$A = \begin{pmatrix} + & - & + \\ a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}$$

$$\det(A) = a_{11} \underbrace{\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}}_{M_{11}} - a_{12} \underbrace{\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}}_{M_{12}} + a_{13} \underbrace{\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}}_{M_{13}}$$

$$= a_{11} M_{11} - a_{12} M_{12} + a_{13} M_{13}$$

$$= a_{ij} M_{ij}$$

$$\det(A) = a_{ij} C_{ij} \quad \begin{matrix} i = \text{fixed} \\ \text{or } j = \text{fixed} \end{matrix}$$

$$C_{ij} = (-1)^{i+j} \underbrace{M_{ij}}_{\text{co-factors}}$$

$$\boxed{\det(A) = a_{ij} C_{ij}} \Rightarrow \text{where } (-1)^{i+j} M_{ij}$$

Let

$$R_1 \rightarrow A = \begin{bmatrix} 3 & 1 & -4 \\ 2 & 5 & 6 \\ 1 & 4 & 8 \end{bmatrix}$$

$$\det(A) = a_{ij} C_{ij}$$

$$= a_{11} C_{11} + a_{12} C_{12} + a_{13} C_{13}$$

$$= 3 \times 16 + 1 \times -10 + (-4) \times 3$$

$$= 48 - 10 - 12$$

$$\checkmark C_{ij} = (-1)^{i+j} M_{ij} \checkmark$$

$$\checkmark C_{11} = (-1)^{1+1} M_{11} = \begin{vmatrix} 5 & 6 \\ 4 & 8 \end{vmatrix} = 40 - 24 = 16 = 26$$

$$\checkmark C_{12} = (-1)^{1+2} M_{12} = - \begin{vmatrix} 2 & 6 \\ 1 & 8 \end{vmatrix} = -(16 - 6) = -10$$

$$C_{13} = (-1)^{1+3} M_{13} = + \begin{vmatrix} 2 & 5 \\ 1 & 4 \end{vmatrix} = +(8 - 5) = 3$$

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$$

$$C_{ij} = (-1)^{i+j} M_{ij}$$

$$\det(A) = a_{ij} C_{ij}$$

$$= a_{ij} C_{ij}$$

$$C_{11} = (-1)^{1+1} M_{11} = 1 \begin{vmatrix} e & f \\ h & i \end{vmatrix} = K$$

$$C_{12} = (-1)^{1+2} M_{12} = -1 \begin{vmatrix} d & f \\ g & i \end{vmatrix} = -K$$

$$= a_{1j} C_{1j}$$

$$= \underbrace{a_{11}}_{\text{circled}} C_{11} + a_{12} C_{12} + a_{13} C_{13}$$

$$= aK + bM + cN$$

$$C_{11} = (-1)^{1+1} M_{11} = 1 \cdot \begin{vmatrix} 2 & 6 \\ 1 & 8 \end{vmatrix} = 10$$

$$C_{12} = (-1)^{1+2} M_{12} = -1 \cdot \begin{vmatrix} 3 & 6 \\ 1 & 8 \end{vmatrix} = -10$$

$$C_{13} = (-1)^{1+3} M_{13} = 1 \cdot \begin{vmatrix} 3 & 5 \\ 1 & 4 \end{vmatrix} = 7$$

Let

$$\det(A) = 26$$

$$A = \begin{bmatrix} 3 & 1 & -4 \\ 2 & 5 & 6 \\ 1 & 4 & 8 \end{bmatrix}$$

$$\det(A) = \underbrace{a_{ij} C_{ij}}$$

$$C_{ij} = (-1)^{i+j} M_{ij}$$

Even
 $i=2$ fix

(Muneeh)

$$-4 \times 3 + 6 \times -11 + 8 \times 13$$

$$= -12 - 66 + 104 = 26$$

$$\det(A) = a_{i3} C_{i3} = a_{13} C_{13} + a_{23} C_{23} + a_{33} C_{33}$$

odd $j=3$

$$C_{13} = (-1)^{1+3} M_{13} = + \begin{vmatrix} 2 & 5 \\ 1 & 4 \end{vmatrix} = 3$$

$$C_{23} = (-1)^{2+3} M_{23} = -1 \begin{vmatrix} 3 & 1 \\ 1 & 4 \end{vmatrix} = -(12-1) = -11$$

$$C_{33} = (-1)^{3+3} M_{33} = + \begin{vmatrix} 3 & 1 \\ 2 & 5 \end{vmatrix} = 13$$

Let

$$A = \begin{bmatrix} 3 & 1 & -4 \\ 2 & 5 & 6 \\ 1 & 4 & 8 \end{bmatrix}$$

$$A = \begin{bmatrix} 3 & 1 & -4 \\ 2 & 5 & 6 \\ 1 & 4 & 8 \end{bmatrix}$$

$$\det(A) = a_{ij} C_{ij}$$

$$= a_{i2} C_{i2}$$

$$\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$$

$$C_{i2} = (-1)^{i+2} M_{i2} = (1)(-10) + (5)(28) + (4)(-26)$$

$$= -10 + 140 - 104 = 26$$

$$C_{12} = (-1)^{1+2} M_{12} = -1 \begin{vmatrix} 2 & 6 \\ 1 & 8 \end{vmatrix} = -(16-6) = -10$$

$$C_{22} = (-1)^{2+2} M_{22} = +1 \begin{vmatrix} 3 & -4 \\ 1 & 8 \end{vmatrix} = +(24+4) = 28$$

$$C_{32} = (-1)^{3+2} M_{32} = -1 \begin{vmatrix} 3 & -4 \\ 2 & 6 \end{vmatrix} = -(18+8) = -26$$

$$\det(A) = a_{ij} C_{ij}$$

($\cdot$) $i = \text{fixed}$
or $j = \text{fixed}$

$$C_{ij} = (-1)^{i+j} M_{ij}$$

$$A = \begin{bmatrix} 2 & -1 & 3 \\ 4 & 0 & 0 \\ -3 & 2 & 5 \end{bmatrix}$$

$$\det(A) = a_{ij} C_{ij}$$

$$= a_{2j} C_{2j}$$

$$= a_{21} C_{21} + \underbrace{a_{22} C_{22} + a_{23} C_{23}}$$

$$= a_{21} C_{21}$$

$$= 4 \times 11 = 44$$

$$C_{ij} = (-1)^{i+j} M_{ij}$$

$$C_{21} = (-1)^{2+1} M_{21}$$

$$= -1 \begin{vmatrix} -1 & 3 \\ 2 & 5 \end{vmatrix} = -(-5-6) = 11$$

$$\Rightarrow \begin{pmatrix} 4 & -2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 9 & 10 & 11 & 12 \\ 7 & 3 & 2 & 1 \end{pmatrix}$$

$$a_{11} C_{11} + a_{12} C_{12} + a_{13} C_{13} + a_{14} C_{14}$$

$$C_{ij} = (-1)^{i+j} M_{ij}$$

$$C_{11} = (-1)^{1+1} M_{11} = \begin{vmatrix} 6 & 7 & 8 \\ 10 & 11 & 12 \\ 3 & 2 & 1 \end{vmatrix}$$

$$C_{12}$$

$$C_{13}$$

$$C_{14}$$

Remark:

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If A is an $n \times n$ triangular matrix (upper triangular, lower triangular, or diagonal), then $\det(A)$ is the product of the entries on the main diagonal of the matrix; that is, $\det(A) = a_{11}a_{22} \cdots a_{nn}$.

$$a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13}$$

① $a_{11}c_{11} + a_{12}c_{12} + a_{13}c_{13}$

$$\rightarrow \begin{bmatrix} 4 & 0 & 0 \\ 3 & 5 & 0 \\ 4 & 0 & -2 \end{bmatrix}$$

upper Triangular diagonal

$$\det(A) = 4 \times 5 \times -2 = -40$$

$$a_{11} \cdot a_{22} \cdot a_{33}$$

②

$$\begin{bmatrix} 4 & 5 & 2 \\ 0 & -2 & 5 \\ 0 & 0 & 3 \end{bmatrix}$$

lower Triangular diag

$$= 4 \times -2 \times 3 = -24$$

③

$$\begin{bmatrix} 5 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

diagonal mat

$$= 5 \times -2 \times 1 = -10$$

$$\begin{pmatrix} 7 & 6 & 7 & 8 & 9 \\ 0 & -2 & 1 & 2 & 3 \\ 0 & 0 & 3 & 4 & 5 \\ 0 & 0 & 0 & 4 & 6 \\ 0 & 0 & 0 & 0 & 8 \end{pmatrix}$$

$$\det(A) = 7 \times -2 \times 3 \times 4 \times 8$$

$$= -14 \times 12 \times 8$$

$$= -1344$$

$$14 \times 12 \times 8 = 1,344$$

$$AB = \begin{pmatrix} 5 & -1 \\ -1 & 10 \end{pmatrix} = 50 - (-1) = 49$$

① $A = \begin{bmatrix} -1 & 2 \\ 3 & 1 \end{bmatrix}$ $B = \begin{bmatrix} -1 & 3 \\ 2 & 1 \end{bmatrix}$ $\det A \cdot \det B$

$$|A| = -7 \quad |B| = -7$$

$$\det(A) + \det(B) = -14$$

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Basic Properties of Determinants

Suppose that A and B are $n \times n$ matrices and k is any scalar. Then:

1. $\det(A + B) \neq \det(A) + \det(B)$
2. $\det(kA) = k^n \det(A)$
3. $\det(AB) = \det(A) \det(B)$
4. A square matrix A is invertible if and only if $\det(A) \neq 0$.
5. $\det(A^{-1}) = \frac{1}{a_{11}} \frac{1}{a_{22}} \cdots \frac{1}{a_{nn}} \rightarrow \text{diagonal}$
6. The points $P_1(x_1, y_1, z_1), P_2(x_2, y_2, z_2), P_3(x_3, y_3, z_3), P_4(x_4, y_4, z_4)$ are not coplanar if and only if the determinant

$$\begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix} \neq 0$$

Remarks:

If A is an $n \times n$ matrix, then the following statements are equivalent.

- (a) A is invertible.
- (b) $Ax = 0$ has only the trivial solution.
- (c) The reduced row echelon form of A is I_n .
- (d) A can be expressed as a product of elementary matrices. *Minor*
- (e) $Ax = b$ is consistent for every $n \times 1$ matrix b .
- (f) $Ax = b$ has exactly one solution for every $n \times 1$ matrix b .
- (g) $\det(A) \neq 0$.

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$A = \begin{pmatrix} -1 & 2 \\ 3 & 1 \end{pmatrix}$$

$$4A = \begin{pmatrix} -4 & 8 \\ 12 & 4 \end{pmatrix}$$

$$= -16 + 96 = -112$$

Property 3 $\det(\underline{ABC}) = \underline{\det A} \times \underline{\det B} \times \underline{\det C}$
 But $\det(A+B+C) \neq \det A + \det B + \det C$

Invertible who has Inverse $\det(A) \neq 0$

If $\det(A) = 0$, matrix is called as Singular matrix

Ex 1 $\begin{pmatrix} 2 & 3 & x \\ 1 & 2 & 1 \\ 4 & 1 & 0 \end{pmatrix}$ if it is Singular find x

$$\det(A) = a_{ij} C_{ij} \Rightarrow a_{23} C_{23}$$

$$C_{ij} = (-1)^{i+j} M_{ij} = \underline{a_{13}} C_{13} + \underline{a_{23}} C_{23} + \underline{a_{33}} C_{33}$$

$$x(-7) + 1(10) = 0$$

$$-7x + 10 = 0$$

$$-7x = -10$$

$$x = \frac{10}{7}$$

$$C_{13} = (-1)^{1+3} M_{13} = \begin{vmatrix} 1 & 2 \\ 4 & 1 \end{vmatrix} = -7$$

$$C_{23} = (-1)^{2+3} M_{23} = - \begin{vmatrix} 2 & 3 \\ 4 & 1 \end{vmatrix} = - (2 - 12) = 10$$

$$C_{33} = (-1)^{3+3} M_{33} = + \begin{vmatrix} 2 & 3 \\ 1 & 2 \end{vmatrix} = 1$$

$$A = \begin{pmatrix} \textcircled{2} & 3 \\ 1 & \textcircled{2} \end{pmatrix}$$

$$|A| = 4 - 3 = 1$$

$$\det(\bar{A}) = \frac{1}{a_{11}} \cdot \frac{1}{a_{22}} = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$$

$$\bar{A}^{-1} = \frac{1}{|A|} \text{adj } A$$

$$A^{-1} = \frac{1}{1} \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}$$

$$A = \begin{pmatrix} 2 & 3 \\ 1 & 2 \end{pmatrix}$$

$$\det(\bar{A}^{-1}) = 4 - 3 = 1$$

$$A = \begin{pmatrix} \textcircled{2} & 3 \\ 1 & \textcircled{2} \end{pmatrix}$$

$$\det(\bar{A}^{-1}) = \frac{1}{a_{11}a_{22}} = \frac{1}{4}$$

$$(1 \quad 2)$$

$$|A| = 4 - 3 = 1$$

$$\text{adj} A = \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}$$

$$\bar{A}^{-1} = \frac{1}{(1)} \begin{pmatrix} 2 & -3 \\ -1 & 2 \end{pmatrix}$$

$$\det(\bar{A}^{-1}) = 4 - 3 = 1$$

$$-\frac{1}{3} \times \frac{1}{4}$$

$$A = \begin{pmatrix} -3 & 4 \\ 2 & 5 \end{pmatrix}$$

$$\det(\bar{A}^{-1}) = -\frac{1}{15}$$

$$|A| = -15 - 8 = -23$$

$$\bar{A}^{-1} = -\frac{1}{23} \begin{pmatrix} 5 & -4 \\ -2 & -3 \end{pmatrix}$$

$$\text{adj} = \begin{pmatrix} 5 & -4 \\ -2 & -3 \end{pmatrix}$$

$$= \begin{pmatrix} -5/23 & 4/23 \\ 2/23 & 3/23 \end{pmatrix}$$

$$= \frac{-15}{(23)^2} - \frac{8}{(23)^2} = \frac{-23}{23^2} = -\frac{1}{23}$$

$$\begin{matrix} \rightarrow 0 \\ Q \\ R \\ S \end{matrix} \begin{vmatrix} x_1 & y_1 & z_1 & 1 \\ x_2 & y_2 & z_2 & 1 \\ x_3 & y_3 & z_3 & 1 \\ x_4 & y_4 & z_4 & 1 \end{vmatrix} = 0 \quad \begin{matrix} P, Q, R, S \\ \text{are Coplanar} \end{matrix}$$

$$AX = b$$

$$n \times 1$$

$$\begin{pmatrix} -2 & 1 & 3 \\ 1 & 2 & 5 \\ 6 & 2 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}$$

$$\text{Compute } \begin{vmatrix} 5 & -7 & 2 & 2 \\ 0 & 3 & 0 & -4 \\ -5 & -8 & 0 & 3 \end{vmatrix}$$

$$\det(A) = a_{23} C_{23}$$

$$= a_{13} C_{13} + a_{23} C_{23} + a_{33} C_{33}$$

Compute $\begin{vmatrix} 0 & 3 & 0 & -4 \\ -5 & -8 & 0 & 3 \\ 0 & 5 & 0 & -6 \end{vmatrix}$

$$= a_{13}C_{13} + a_{23}C_{23} + a_{33}C_{33} + a_{43}C_{43}$$

$$= a_{13}C_{13}$$

$$(2)(10) = 20$$

$$C_{13} = (-1)^{1+3} M_{13}$$

$$= \begin{vmatrix} 0 & 3 & -4 \\ -5 & -8 & 3 \\ 0 & 5 & -6 \end{vmatrix}$$

$$a_{i1}C_{i1}$$

$$= a_{11}C_{11} + a_{21}C_{21} + a_{31}C_{31}$$

$$= a_{21}C_{21}$$

$$C_{13}$$

$$= (-5)(-2)$$

$$= 10$$

$$C_{21} = (-1)^{2+1} M_{21} = -1 \begin{vmatrix} 3 & -4 \\ 5 & -6 \end{vmatrix}$$

$$= -(-18 + 20)$$

$$= -2$$